

# Controls / Optimization HW 1

Sina Hazeghi

September 3, 2026

## 1) Discrete System with Delay

System:  $x_{t+1} = Ax_t + Bu_t$ , with control law  $u_t = Kx_t$ . Stable iff  $\rho(A + BK) < 1$ , where  $\rho(M) = \max\{|\lambda| : \lambda \text{ eigenvalue of } M\}$ .

Since  $|\lambda| = \sqrt{a^2 + b^2}$  for  $\lambda = a + bi$ , we always have  $|\lambda| \geq 0$ , so  $\rho(A + BK) < 0$  is never possible.

### a) Delayed feedback

Consider delayed feedback law  $u_t = Kx_{t-1}$ . What is sufficient for stability?

$$x_{t+1} = Ax_t + Bu_t = Ax_t + BKx_{t-1}$$

Let the new state be  $z_t = \begin{pmatrix} x_t \\ x_{t-1} \end{pmatrix}$ . Then

$$z_{t+1} = \begin{pmatrix} x_{t+1} \\ x_t \end{pmatrix} = \begin{pmatrix} Ax_t + BKx_{t-1} \\ x_t \end{pmatrix} = \begin{pmatrix} A & BK \\ I & 0 \end{pmatrix} \begin{pmatrix} x_t \\ x_{t-1} \end{pmatrix}$$

Let  $M = \begin{pmatrix} A & BK \\ I & 0 \end{pmatrix}$ . Then  $\rho(M) < 1$  is necessary and sufficient for stability of the system.

### b) Two-step delayed feedback

Now let  $u_t = Kx_{t-2}$ . Let

$$z_t = \begin{pmatrix} x_t \\ x_{t-1} \\ x_{t-2} \end{pmatrix}, \quad z_{t+1} = \begin{pmatrix} Ax_t + BKx_{t-2} \\ x_t \\ x_{t-1} \end{pmatrix}$$

So

$$z_{t+1} = \begin{pmatrix} A & 0 & BK \\ I & 0 & 0 \\ 0 & I & 0 \end{pmatrix} z_t = M' z_t$$

So necessary/sufficient for  $\rho(M') < 1$

### c) Concrete Example

$$x_{t+1} = \begin{pmatrix} 0.5 & 0 \\ 0 & 1 \end{pmatrix} x_t + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u_t$$

Find  $K$  such that  $u_t = Kx_t$  is stable, while  $u_t = Kx_{t-1}$  is unstable.

Want:  $\rho(A + BK) < 1$  and  $\rho(M) > 1$ , where  $M = \begin{pmatrix} A & BK \\ I & 0 \end{pmatrix}$ .

Let  $K = (K_1 \ K_2)$ . Then we have undelayed dynamics:

$$A + BK = \begin{pmatrix} 0.5 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} (K_1 \ K_2) = \begin{pmatrix} 0.5 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ K_1 & K_2 \end{pmatrix} = \begin{pmatrix} 0.5 & 0 \\ K_1 & 1 + K_2 \end{pmatrix}$$

For delayed dynamics let  $x_{t+1} = Ax_t + BKx_{t-1}$ :

$$x_{t+1} = \begin{pmatrix} \frac{1}{2}x_t^{(1)} + 0x_{t-1}^{(1)} + 0x_{t-1}^{(2)} \\ x_t^{(2)} + K_1x_{t-1}^{(1)} + K_2x_{t-1}^{(2)} \end{pmatrix}$$

**Key info:** In the first case,

$$A + BK = \begin{pmatrix} 0.5 & 0 \\ K_1 & 1 + K_2 \end{pmatrix}, \quad \text{eigenvalues} = \{0.5, 1 + K_2\}$$

since it's triangular. So  $K_1$  can be any number without affecting the stability of the (undelayed) system. Need  $K_2 \in (-2, 0)$  to satisfy stability.

For the delayed system, we then see

$$x_{t+1} = \begin{pmatrix} \frac{1}{2}x_t^{(1)} \\ x_t^{(2)} + K_1x_{t-1}^{(1)} + K_2x_{t-1}^{(2)} \end{pmatrix}$$

The  $K_1$  term will go to 0 since  $x_t^{(1)} \rightarrow 0$  as  $t \rightarrow \infty$ . Let  $x_t^{(2)}$  be  $y_t$  for naming convenience; the recurrence equation is:

$$y_{t+1} = y_t + K_2y_{t-1}$$

For linear discrete systems we know components generally grow by a weighted sum of scaling factors  $\lambda_i$  at each timestep (recall the general solution  $x[k] = M^k x[0]$ ). We know only 1  $\lambda$  matters here since the other component of  $x_t$  goes to 0. So it's reasonable to solve for it by plugging  $y_t = \lambda^t$ :

$$\lambda^{t+1} = \lambda^{t-1}(\lambda + K_2) \rightarrow \lambda^2 - \lambda - K_2 = 0$$

So we need some  $K_2$  such that the equation above has  $|\lambda| > 1$ ; try  $K_2 = -1.25$ , which satisfies both constraints.

So  $K = (0 \ -1.25)$  satisfies both undelayed stable + delayed unstable.

## 2) Uncertain Pendulum

Inverted pendulum with damping, described by the system:

$$\dot{x} = \begin{pmatrix} \dot{\theta} \\ \ddot{\theta} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ \omega^2 & -\gamma \end{pmatrix} x + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u$$

We want to determine  $K = (K_1 \ K_2)$  to stabilize the system for  $\omega^2 \in [5, 10]$  and  $\gamma \in [0, 1]$ . What set of  $(K_1, K_2)$  works?

$$A + BK = \begin{pmatrix} 0 & 1 \\ \omega^2 & -\gamma \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ K_1 & K_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ \omega^2 + K_1 & K_2 - \gamma \end{pmatrix}$$

Characteristic equation:

$$\det(A + BK - \lambda I) = 0 \implies \det \begin{pmatrix} -\lambda & 1 \\ \omega^2 + K_1 & K_2 - \gamma - \lambda \end{pmatrix} = 0$$

$$\lambda^2 + (\gamma - K_2)\lambda + (\omega^2 + K_1) = 0$$

$$\lambda = \frac{(K_2 - \gamma) \pm \sqrt{(\gamma - K_2)^2 - 4(\omega^2 + K_1)}}{2}$$

Stable if  $\text{Re}(\lambda) < 0$ .

In general, for  $x^2 + bx + c = 0$ :  $\text{Re}(x) < 0$  iff  $b > 0$  and  $c > 0$ .

Here  $b = \gamma - K_2$  and  $c = \omega^2 + K_1$ . For stability over the *entire* range  $\omega^2 \in [5, 10]$ ,  $\gamma \in [0, 1]$ , we need the worst case (smallest values) to still satisfy the inequalities:

$$\begin{aligned} \gamma - K_2 > 0 \quad \forall \gamma \in [0, 1] &\implies K_2 < 0 \\ \omega^2 + K_1 > 0 \quad \forall \omega^2 \in [5, 10] &\implies K_1 > -5 \end{aligned}$$

**So the stabilizing set is:**  $K_1 > -5$  and  $K_2 < 0$ .

### 3) Exponential Stability and Friction

We have a block with friction, dynamics:

$$\ddot{x} = \begin{cases} 0, & \dot{x} = 0 \text{ and } |u| \leq \mu \quad (\text{sticking}) \\ u - \mu, & \dot{x} > 0 \text{ or } (\dot{x} = 0 \text{ and } u > \mu) \quad (\text{sliding}) \\ u + \mu, & \text{otherwise} \end{cases}$$

Prove there do not exist gains such that the control law  $u = -k_p x - k_d \dot{x}$  is exponentially stable about the origin.

1. Exponential stability says the system is asymptotically stable AND convergence is bounded by an exponential function.
2. Asymptotic stability says Lyapunov stable AND:  $\exists \delta > 0$  s.t.  $\|x(0)\| < \delta \implies x(t) \rightarrow 0$  as  $t \rightarrow \infty$

Argue that this system is not asymptotically stable:

Suppose it is asymptotically stable to the origin; then there's a  $\delta$ -ball s.t. if we start there we will converge to the origin as  $t \rightarrow \infty$ .

Let  $\dot{x}(0) = 0$  be the initial condition. Then by our control law:  $u(0) = -k_p x(0) - k_d \dot{x}(0) = -k_p x(0)$ .

Notice then that for any  $\mu > 0$ :

$$|u| = |k_p| \cdot |x(0)| \leq \mu$$

If  $k_p = 0$  then  $u = 0 < \mu$ . Otherwise for nonzero  $k_p$ , no matter how large  $|k_p|$  is we can always find  $|x(0)| < \min\left(\frac{\mu}{|k_p|}, \delta\right)$ ,  $x(0) \neq 0$ , within the  $\delta$ -ball s.t.  $\ddot{x} = 0$ ; and since  $\dot{x}(0) = 0$  as well, our trajectory is simply  $x(t) = x(0)$ , and it won't converge to the origin. Thus we can conclude that this system is not asymptotically stable, and thus it's not exponentially stable.

## 4) Gradient Flow

Let the dynamical system be the negative gradient of  $l : \mathbb{R}^n \rightarrow \mathbb{R}$ :

$$\dot{x} = -\nabla l(x)$$

### a) Does strongly convex $\implies$ stable equilibrium?

We know that an equilibrium is locally exponentially stable iff the linearization is stable. The linearization of the system is:

$$\dot{x} = -\nabla l(x) \approx -H(x)x$$

where  $H(x) = \nabla^2 l(x)$ . Since  $l$  is strongly convex, we know the minimum eigenvalue of  $H$  is  $> \sigma$ ; therefore we can conclude the maximum eigenvalue of  $-H$  is  $< -\sigma < 0$ , so this system is locally exponentially stable since all eigenvalues of its linearization satisfy  $\text{Re}(\lambda) < 0$ .

### b) Gradient Descent

Consider  $l(x) = x^T Q x$  for  $Q \succeq \sigma I$ ,  $\sigma > 0$ . Discretizing the dynamics in gradient descent:

$$x_{t+1} = x_t - \Delta t \nabla l(x_t) = (I - 2\Delta t Q)x_t$$

Then we only need to show  $\rho(I - 2\Delta t Q) < 1$ .

$$Q \succeq \sigma I \rightarrow -2\Delta t Q \preceq -2\Delta t \sigma I \implies I - 2\Delta t Q \preceq (1 - 2\sigma \Delta t)I$$

So now we can bound the eigenvalues of our dynamics:

$$\rho(I - 2\Delta t Q) \leq \rho((1 - 2\Delta t \sigma)I)$$

Since all of  $I$ 's eigenvalues are 1, it's sufficient to have:

$$|1 - 2\Delta t \sigma| < 1$$

which gives us the bound  $0 < \Delta t < \frac{1}{\sigma}$ . So timesteps within this bound will make our system globally stable (since it's linear).

## 5) 2D Equilibrium

### a) Equilibrium Points

$$f(x) = 0 \text{ for points: } x^* = \{(-1, 2)^T, (-1 - \sqrt{3}, 2)^T, (-1 + \sqrt{3}, 2)^T\}$$

### b) Stable Equilibrium

To see if our equilibrium converges we define  $z = x - x^*$ ; then we have the system in terms of  $z$ :

$$\dot{z} = f(z + x^*) \approx f(x^*) + J_f(x^*)z = J_f(x^*)z$$

So we can check if equilibrium points are stable by linearizing as shown above and checking the eigenvalues of the Jacobian of  $f(x)$ .

$$J_f(x) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 + (x_1 + 1)^2 & -1 \end{pmatrix}$$

And evaluate it at all of our equilibrium points.

For  $x^* = (-1, 2)^T$  we get:

$$J_f(x^*) = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}$$

This has eigenvalues which both satisfy  $\text{Re}(\lambda) < 0$  ( $\text{Re}(\lambda) = -0.5$ ), so this equilibrium is locally exponentially stable.

For the other 2 fixed points, linearization gives us eigenvalues  $\lambda = \{1, -2\}$ ; these fixed points are not stable (see exact Jacobians / work in attached Jupyter notebook).

### c) Phase Plot

Please refer to the attached Jupyter notebook for details; we compute the 2D phase plot:

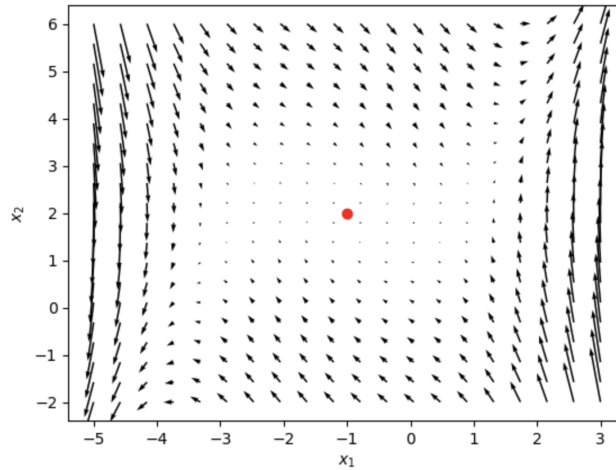


Figure 1: 2D Phase Plot around Equilibrium Point  $(-1, 2)$

### d) Approximating Region of Attraction

To approximate the region of attraction we can sample points on our grid and integrate forward for a fixed time horizon  $T$ .

For points whose final trajectory is close to the equilibrium, we say they are part of the region of attraction, and we use scipy's `ConvexHull` function to draw a convex shape about this region to demarcate it. Code in the attached Jupyter notebook.

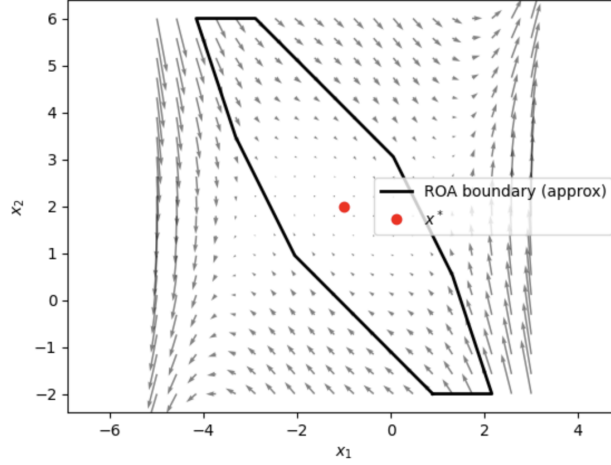


Figure 2: Phase Plot with Approx. RoA boundary

### e) Converging/Diverging Trajectories

We sample 2 initial conditions, one diverging and one converging, and plot:

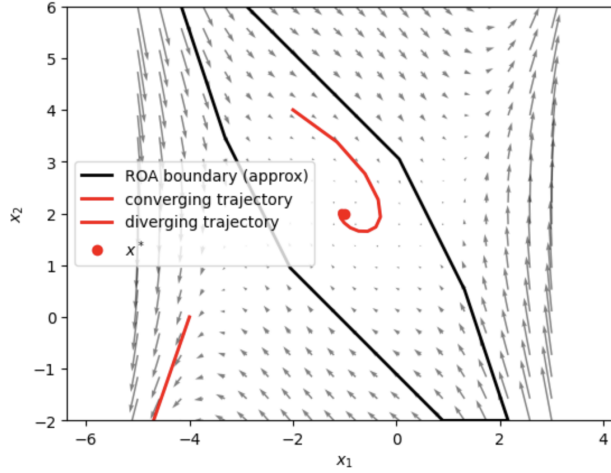


Figure 3: Phase Plot with Sampled Trajectories

## 6) 1D Phase Plot

We have system

$$\dot{x} = x^2 + a$$

for  $a \in \mathbb{R}$  constant.

### a) 1D Plot with $a = -1$

Equilibrium for  $\dot{x} = x^2 - 1$  is just  $x = \pm 1$ . In general for any  $a$ ,  $\dot{x} = 0$  for  $x = \pm\sqrt{-a}$ .

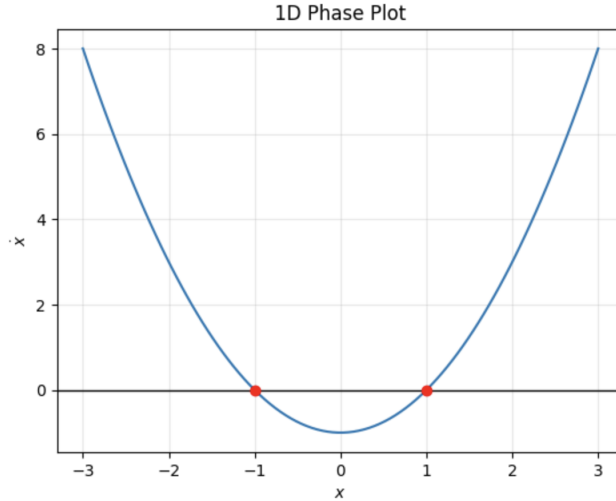


Figure 4: 1D Phase Plot of  $\dot{x} = x^2 - 1$

### b) Bifurcation diagram

For the bifurcation diagram we are plotting the equilibrium points  $x^*$  on the y-axis and  $a$  along the x-axis.

We take  $\dot{x} = x^2 + a$ , set it to 0, and see that  $x^* = \pm\sqrt{-a}$ .

### Stability / Equilibrium

We can see which of the fixed points of the system are stable by linearizing  $f(x)$ :

$$df/dx = 2x$$

For our equilibrium points  $x^* = \pm\sqrt{-a}$ :

- Case 1:  $a > 0$ , then  $x^*$  is imaginary, no real equilibrium point, so nothing to say about stability.
- Case 2:  $a = 0$ , then  $x^* = 0$  and the linearized dynamics are also 0, so we can't say whether stable/unstable.
- Case 3:  $a < 0$ , then we have 2 real fixed points:  $\sqrt{-a}$  unstable and  $-\sqrt{-a}$  locally exponentially stable.

### c) Derive Region of Attraction

Assuming a stable equilibrium exists (Case 3), we can derive an expression for the region of attraction by examining the behavior of  $\dot{x}$  in the different regions delineated by the 2 real fixed points of the system.

1. For  $x < x_1^* = -\sqrt{-a}$  we know  $\dot{x} > 0$ , so  $x \rightarrow -\sqrt{-a}$ .
2. For  $x_1^* < x < x_2^* = \sqrt{-a}$  we know  $\dot{x} < 0$ , so here also  $x \rightarrow -\sqrt{-a}$ .

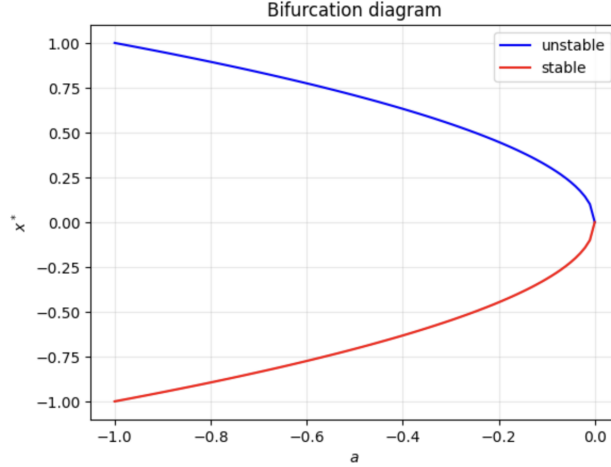


Figure 5: Bifurcation Diagram

3. For  $x > x_2^*$  we have  $\dot{x} > 0$  and  $x \rightarrow \infty$ .

So the region of attraction for  $x \rightarrow x_1^*$  is the set of initial conditions  $x(0) \in (-\infty, \sqrt{-a})$ , and for  $x \rightarrow x_2^*$  it is only  $x(0) = \sqrt{-a}$ .

## 7) Stability of Nonlinear System

We have system  $x = (r, \theta)^T$  with nonlinear dynamics

$$\dot{x} = f(x) = \begin{pmatrix} -r + \theta + 1 \\ \sin^2(\theta/2) \end{pmatrix}$$

a)  $x = (1, 0)^T$

Plugging this  $x$  into  $f(x)$  we get

$$f((1, 0)^T) = \begin{pmatrix} -1 + 0 + 1 \\ \sin^2(0) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

so yes, this is an equilibrium point.

b) Is  $x$  locally asymptotically stable?

We compute the Jacobian of the dynamics:

$$J_f(x) = \begin{pmatrix} -1 & 1 \\ 0 & \sin(\theta/2) \cos(\theta/2) \end{pmatrix}$$

And notice that when we plug in our equilibrium point into this matrix we have

$$J_f(x^*) = \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix}$$

which has eigenvalues  $\lambda = \{-1, 0\}$ , so our test is inconclusive; not sure if this is locally exponentially stable (and we don't have a different test for local asymptotic stability of a nonlinear system).

### c) The Nonlinear Dynamics in the Limit

For any initial condition we want to know what happens to  $\lim_{t \rightarrow \infty} x(t)$ .

For the first component  $x_1 = r$ , this depends on both  $r$  and  $\theta$ ; however, we know that  $\theta(t) \in [-\pi, \pi)$  and it only depends on itself:

$$\dot{\theta} = \sin^2(\theta/2)$$

Notice that  $\dot{\theta} \geq 0$  for any value of  $\theta$ , and that  $\dot{\theta} = 0$  only for  $\theta = 0$ .

Therefore, no matter what  $\theta(0)$  is, it will always increase (mod  $2\pi$ ) until it reaches  $\theta = 0$ ,  $\dot{\theta} = 0$ , at which point it will stop.

Now that we know  $\theta = 0$  in the limit, we can reason that  $\lim \dot{r}(t) = -r + 0 + 1$ , so for any  $r < 1$  we have  $\dot{r} > 0$ , and for any  $r > 1$  we have  $\dot{r} < 0$ , so the radius will also always approach  $r = 1$ .

So we can conclude that for any initial condition,  $\lim_{t \rightarrow \infty} x(t) = (1, 0)^T$ .